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(Key and Solutions for AIMCAT2407)

Key

SECTION – I

1.	C	7.	B	13.	A	19.	B
2.	A	8.	C	14.	D	20.	2134
3.	D	9.	A	15.	C	21.	D
4.	B	10.	D	16.	C	22.	D
5.	D	11.	A	17.	2314	23.	D
6.	C	12.	B	18.	C	24.	2413

SECTION – II

1.	C	7.	D	13.	C	19.	D
2.	C	8.	D	14.	A	20.	B
3.	D	9.	B	15.	D	21.	120
4.	A	10.	50	16.	128	22.	30
5.	B	11.	90	17.	3		
6.	D	12.	D	18.	B		

SECTION – III

1.	C	7.	2	13.	D	19.	A
2.	4	8.	A	14.	B	20.	D
3.	C	9.	56	15.	9	21.	807040
4.	D	10.	D	16.	B	22.	C
5.	42924	11.	B	17.	16	23.	A
6.	A	12.	C	18.	B	24.	70

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 562

1. The disappearance of the five aircraft (Flight 19) without any evidence of a crash or wreckage led to fanciful hypotheses as can be gathered from the penultimate paragraph of the passage. The author tries to suppress the sensational legends that have grown around the Bermuda Triangle by providing suitable evidence in the last paragraph. (From the first part of the last para, which presents a view not attributed to anyone else, one can see that the author is in favour of the rational-explanation theory.) Hence, the author is trying to initiate a debate about a well-known phenomenon. This makes choice C the correct answer.
Option A: The author does not report new research findings. This makes choice A wrong.
Option B: The evidence presented in the passage only raises a question against previously held views. It does not invalidate them since the author leaves the issue a little open-ended. Choice B is not the answer.
Option D: The author does not explain the actual causes behind each of the mysterious occurrences. He attempts to explain that the legends surrounding the Bermuda Triangle are doubtful at best. So choice D is not correct.

Choice (C)

2. The word 'contemporaneous' means existing at or occurring in the same period of time.
Hence choice A is the answer. The remaining choices do not apply.

Choice (A)

3. Option A: The third para begins by saying: The legend of the Bermuda Triangle will be forever tied to Flight 19. But in the first para, we are told that the Bermuda Triangle did not receive its famous nickname until 1964. Choice A is not true.

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Option B: Refer to the penultimate para of the passage. The disappearances have been attributed to the machinations of enormous sea monsters, giant squid, extra-terrestrials, alien abductions, Atlantis, time warps, reverse gravity fields, and ocean flatulence. The Tempest is not a probable causal factor of the disappearance of Flight 19. “The Tempest” was William Shakespeare’s play which was based on a real-life Bermuda shipwreck and which may have enhanced the area’s aura of mystery. Choice B is not true.

Option C: When all the facts are laid out, the tale of Flight 19 becomes less puzzling. From there on, the author gives valid reasons for the crash and the mysterious disappearance of the aircraft and the crew. Choice C contradicts this, and thus it can be ruled out.

Option D: From the penultimate para of the passage, we can infer that choice D is correct.

Choice (D)

4. Option A: Choice A seems to imply that mysterious disasters are not merely limited to the Bermuda triangle. But choice A is neither given nor implied in the passage, especially the last paragraph.

   Option B: “We might then just as well” is a hypothetical deduction that the author makes in the last paragraph after he mentions his reasoning: Most of the associated mishaps can be explained by rational means and many of them did not occur within the Bermuda Triangle. Sea disasters as distant as Portugal, Ireland and the Pacific and Indian Oceans have been blamed on the Bermuda Triangle. The author was being sarcastic when he states that we should rename the Bermuda Triangle. Choice B can be inferred.

   Option C: Choice C is incorrect as the author begins the last para with “There are some obstacles in taking the Bermuda Triangle legend seriously” Also choice C does not take into consideration the ‘renaming’ of the Bermuda triangle as ‘The Worldwide Curse of All Seas’.

   Option D: We might then just as well rename it as “The Worldwide Curse of All Seas”. **Some have turned this fact on its head** by proposing this as evidence that the Devil’s Triangle is expanding in scope. But the author does not believe that the Bermuda Triangle causes mysterious disasters. Hence, choice D is not correct.

   Choice (B)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 553

It wasn’t to draw a contrast with the first insight at all. Option A is not the answer.

Option B: The author has clearly explained that it isn’t incorrect to say that hypertension is a genetic disease. There is no gene for hypertension doesn’t negate that statement because as we can understand, it is caused by the influence of not one gene, but several together. So, the option misrepresents information. It is not the answer.

Option C: The para doesn’t generalise that physiology is always controlled by several genes. It could be one and it could be several. Either way, this particular figure of speech was just about hypertension, and not about all aspects of physiology. Option C is not the answer.

Option D: The expression compares several genes to the strings tugging a puppet’s arms to create what we call hypertension. So, blood pressure isn’t controlled by one specific gene but by several genes. That is what the author was trying to explain figuratively. Option D is the answer.

Choice (D)

6. Option A: This can be understood from the examples. The author clearly mentions that ‘*For some disorders, such as Tay-Sachs disease, penetrance was largely complete*’. The same is not true for breast cancer, where the penetrance is not complete and not all women with the mutation will develop the issue. So, Option A can be understood. Option A is not the answer.

   Option B: The example of breast cancer helps us understand that one may have a mutation while not developing the disorder because of incomplete penetrance. So, inheritance of a gene mutation doesn’t really translate all the time to the disorder. So, it is not dependent on inheritance alone. So, this option can be understood. Option B is not the answer.

   Option C: We cannot be sure of this. We know a few gene mutations that virtually guarantee the disorder and a few that do not guarantee the disorder. But, we cannot understand from these two examples that it is possible to look at ***most*** gene mutations and predict whether the disorder is guaranteed or not. From the passage, we cannot say whether most of the gene mutations connected to disorders have been studied and understood. So, it is quite likely we come across many mutations where we are not sure about the level of penetrance. Option C is the answer.

   Option D: Breast cancer gives a good example of this. There are gene variants, which even when inherited, may not be able to penetrate into the physical or morphological features because of incomplete penetrance. Option D is not the answer.

   Choice (C)

5. Option A: The author in using the expression of a tangle of strings isn't really trying to compare the first and second insights of McKusick. This was a figure of speech purely to explain what was happening with blood pressure, which in turn, was an example for McKusick's second insight.

7. Option A: The author mentions about mutations being statistical not because the discussion is centred on whether mutations guarantee disorders or not. (That depends on penetrance, discussed earlier in the passage). The discussion here is about whether a mutation on its own causes a disease. So, Option A while being a true statement is the wrong reason for why the author expressed the remark 'statistical rather than normative concept' about mutations. Option A is not the answer.

Option B: This is what the author was trying to explain as can also be understood from '*By itself, then, a mutant, or a mutation, can provide no real information about a disease or disorder.*' The mutation doesn't carry the disease and it does not imply '*a gain or loss of function*'. It only leads to a mismatch. The mismatch causes the disease. That's what the author was trying to highlight. Option B is the answer.

Option C: Environments don't cause diseases either. The author specifies that neither mutations nor environments cause diseases. It is the mismatch that leads to disorders. Either way this option doesn't explain what the author meant by calling a mutation a statistical concept. Option C is not the answer.

Option D: The discussion here is not about whether mutations can be inherited, but why mutations aren't really responsible directly for the diseases. This option doesn't explain why the author mentioned that mutations are a statistical concept. His comment has got nothing to do with inheritance. Option D is not the answer.

Choice (B)

8. Option A: The fourth insight of the passage suggests that mutations do not offer information about a disease. A mutation is just a variation and it is the mismatch or incongruity with the environment that leads to the disease. So, this research isn't really consistent with the ideas presented in the passage. Option A is not the answer.

Option B: The author has already specified that

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 544

9. The author mentions that when we experience conflict, we tend to come up with more original and creative solutions than when we are in a cooperative mood. A situation where there is an introduction of conflict in decision-making process to bring about a positive, creative solution will be the answer to this question.

Option A: A devil's advocate is a person who expresses a contentious opinion in order to provoke debate or test the strength of the opposing arguments. Bringing in such a person helps with introducing conflict into a situation, thereby promoting a deep exploration of alternatives and options. This is the correct answer.

Option B: Hiring people from diverse backgrounds does not guarantee a difference of opinion. While there could be fresh perspectives, this does not necessarily imply that there will be a conflict. This option is incorrect.

Option C: Having a pressing problem implies that there is a crucial problem that needs to be solved immediately. The author mentions that rebels don't adhere to deadlines. Therefore, solving a pressing problem using a team full of rebels, however creative they may be, is not a great idea. This is not the correct answer.

Option D: The main objective of setting conflicting goals, according to the author, is not to incite disagreements. In the example the author uses, conflicting goals were set to improve creative thinking. We do not know if these conflicting ideas actually caused people to disagree or to work together. This is not the answer. Choice (A)

incomplete penetrance sometimes results in a gene not causing a disorder/disease despite being inherited. The passage doesn't make a connection between inheritability and penetrance at all. A gene could be inherited and may not have penetrance. In fact, the question of penetrance arises only if the gene is inherited. So, Option B is not the answer.

Option C: We know that incomplete penetrance leads to the non-manifestation of a disease at times, even though the gene has been inherited. So, a research into what causes incomplete penetrance or what leads to the lack of expressivity of a gene could help in understanding genes better. Option C is, therefore, the answer.

Option D: This doesn't add any value to the arguments or theory presented in the passage. Hyper-functionality itself has been mentioned as an example of one of the unexpected effects of a gene mutation/mismatch between the genetic endowment and one's environment. However, the effect of environment on hyper-functionality is not necessary for the ideas discussed in the passage. Option D is not the answer. Choice (C)

10. Option A: The author mentions that rebels "understand that it's always worth resisting time pressure to give yourself a moment to reflect" and also that they are "prone to disagreement", which is a positive thing since research finds that with conflict "we generate more original solutions than when we are in a more cooperative mood". This inference is something the author will agree with and is not the answer.

Option B: The author mentions that rebels think outside the box and come up with creative solutions, even if, in the process, they delay timelines and cause conflict. The author cites two examples, one which is the could vs should challenge and the setting of conflicting goals, both of which pushed the people to give innovative solutions. This is a correct inference and is not the answer.

Option C: There is a clear distinction between rulebreakers who cause trouble and give no end result and rulebreakers who contribute. The former is considered as troublemakers and the latter as rebels. The author believes that these rebels are worth emulating for payoff they give. This is a correct inference and is not the answer.

Option D: In the first para of the passage, the author mentions that there are people who break rules and cause trouble by creating conflict and delaying timelines without any positive end result. Rebels, however, offer creative solutions through the same process. This is what differentiates the troublemakers from the rebels. However, in the option, it states that the "creativity" of solution they offer differs. However, the author states that in the case of troublemakers, there is no solution (creative or otherwise) at all. So, the author is not likely to agree with this inference, making it the correct answer. Choice (D)

11. The author's argument in the last para of the passage is that we can all learn from a leader, who is a rebel, who not only influences his employees, but also creates an environment where other rebels(employees) are encouraged to grow and be creative. The question asks us to find the statement, which when made false, invalidates the author's conclusion. The way to go about this is to consider each statement as false and then check if this

liberty to think creatively and come up with innovative solutions. The author has not conveyed this message through the passage. This is the correct answer.

Option C: The author cites the example of the head chef and owner of Osteria Francescana to emphasize on the fact that leaders should set an example and encourage deviating from the norm for innovative solutions. This is a correct message and is not the answer.

Option D: In difficult ethical challenges, the author believes that the "should do" mindset "gets us stuck on the trade-off", whereas in the "could do" approach, we stay open minded. This is a message the author intends to convey and is not the answer. Choice (B)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 523

negation contradicts the author's stance.

Option A: This option when made false says that the attitude displayed by leaders in not important in shaping the attitude of the employees. This goes directly against the author's argument that leaders do shape their employees as well as the workplace. This is the correct answer.

Option B: On considering this option as false, we see that it says that employees should get less leeway than managers. The last para does not compare the leeway given to employees and managers. The author recommends that an environment where people are allowed to be creative is something that we can all learn from. This option does not contradict the author's argument, as it is not related to his conclusion.

Option C: Considering this statement as false adds weight to the author's argument that a leader and a supportive environment that makes a rebel feel at home. It does not contradict the author's argument and is not the answer.

Option D: Proving this statement false would mean that rulebreakers do not exploit the leeway given to them – they either do not use it or they use it in a good way. This adds weight to the author's argument. This option is not the answer.

Choice (A)

12. Option A: While discussing the concept of conflict at the workplace, the author states that experiencing conflict leads to better, more innovative solutions - "When there is tension, we also tend to scrutinize options and deeply explore alternatives, which leads to novel insights". The author also mentions that "conflict and disagreements can be taken too far". From these lines, we see that the author encourages conflict (in moderation) for creative solutions. This is not the answer.

Option B: The author endorses the opposite of setting constraints on personal initiative, i.e., the author believes that people should be given the

13. In the penultimate para, the author mentions that the transhumanist movement receives a lot of support from California and Silicon Valley, which are rich regions. He also says that only the rich will be able to afford to enhance themselves, which implies that transhuman technology is expensive and the poor cannot afford it. This will result in further economic divide in such regions.

Option A: We can infer that transhuman technology is expensive since "...only some people will be able to afford to become enhanced while many others lose out" which means that not everyone will have access to it. Hence, Option A is the correct answer.

Option B: While it is true that transhuman technology has received support from California and Silicon Valley, it is incorrect to say that it has received "most" of its support from there. The passage mentions that rich regions support this movement and these regions could be across the globe. Hence, Option B is incorrect

Option C: This option is too strong in stating that "*there is **great** economic divide*" and "*the divide **will** increase*" which does not have to be the case. The author simply states that there is economic disparity which could increase because of this technology. Again, this is directly stated in the para and does not have to be inferred at all. Hence, Option C is incorrect.

Option D: Critics of transhuman technology talk about only the rich being able to afford to enhance themselves and the poor cannot afford this technology. However, we cannot say this is the only reason why critics disapprove of transhuman technology. Hence Option D is not the answer.

Choice (A)

14. The author uses the example of the "powered clothing" to illustrate how "technological or medical advances" are helping the "injured, sick or elderly". In that case, the powered clothing is helping the

elderly by acting as an exoskeleton and providing them the support required to stand up and move around. This is something the author approves of and considers as ethical. Then he mentions that

humans and genuinely intelligent computers and AI systems". Hence, Option A is incorrect.

Option B: This option does not fully capture the author's stance. While it is true that he endorses this technology is "…then adopted by the healthy or young to boost their lifestyle or performance" and this he disapproves of.

Option A: This is an example of how a young, healthy person uses transhuman technology to improve their health. In this case, Amy is healthy, but she has a stent installed, but she does not have weak arteries. She has a stent installed, just in case. Option A is not the answer.

Option B: This is also an example of how technology is used by the young and healthy to boost their lifestyle and performance. Here, Jake has the natural lens of his eyes replaced with artificial ones to boost his performance at work. There is no mention as to whether the natural lens is damaged, which means that he has probably substituted a healthy human organ with an artificial one. Hence, Option B is incorrect.

Option C: While Lizzie is an elderly person, in her seventies, she undergoes a procedure to improve the appearance of her skin. This is an aesthetic and not a functional change. It is an example of how transhuman technology is used to boost one's lifestyle. Option C is not the answer.

Option D: While Mikhail is still young, he has lost his limbs in the war. His choice of having the carbon-fibre blades as limbs is out of necessity. This is an example of the "...injured, sick or elderly" using transhuman technology and is similar to the example of the "powered clothing". Option D is the answer. Choice (D)

15. From the passage, we gather that the author is not completely against the idea of using technology to augment human experiences. He talks about how transhumanism has led to the invention of "wooden legs, hearing aids, spectacles and false teeth" and also sounds positive about further innovation that could lead to humans having vastly increased intelligence, strength, and lifespans-"a near embodiment of gods". He also mentions that using technology such as "powered clothing" to help the injured, sick and elderly is a good thing. What he disapproves of is the transhumanists taking this to the extreme and using it to enhance "their lifestyle or performance". We see that he questions the ethicality of replacing perfectly good human body parts with transhuman technology, for personal gain. He uses the example of carbon-fibre blades in athletes to emphasize this. He also talks about how it could impact the society by widening the economic gap.

Option A: While the first part of the option, about the author being on the fence about the ethicality of transhumanism is true, the second part, where it states that he is leaning towards transhumanism and its benefits, is incorrect. In fact, the opposite is happening. He mentions how transhumanists take it to the extreme when they talk about how "Biotechnology will enable a union between transhuman technology as beneficial for the injures, sick and elderly, this is a very small part of the passage and does not account for the author's stance throughout the passage. Option B is not the answer.

Option C: This option is succinct and correctly captures the author's stance in the passage. While the author is optimistic about how it could help people in need, he is also worried about what impact transhumanism will have, when taken to the extreme. We see his dilemma when he asks, "Is that a desirable goal?" and "...when the technology reaches this level, will it be ethical to allow surgeons to replace someone's limbs with carbon-fibre blades just so they can win gold medals?". He also seems relieved that such extreme forms of transhuman technology will take decades to reach fruition, as can be gathered from the last para. Hence, Option C is the correct answer.

Option D: This option is correct when it says that the author suggests being cautious about the future applications and impact of transhumanism when taken to the extreme. However, it does not talk about the author being optimistic about its uses for the injured, sick and the elderly. Hence, option D is not complete and cannot be the answer. Choice (C)

16. From the para "Transhumanists believe that..." we understand that transhumanists are very much in favour of the marriage of the human body and AI system. "Enhancing" human beings by replacing human organs with transhuman technology, irrespective of whether this is needed or not is something that they strongly advocate.

Option A: Transhumanists believe that technology can free humans from the shackles which are "the frailties of the human body", by replacing failing organs with transhuman devices. Hence, this can be inferred. Option A is not the answer.

Option B: While the author is on the fence about the ethicality of replacing perfectly healthy human body parts with transhuman technology, we gather that "[advocates] of transhumanism" endorse going beyond "natural barriers and limitations that constitute an ordinary human being". To support this, the author gives an example of how some athletes might prefer carbon-fibre blades instead of natural legs, "just so they can win gold medals". Hence Option B is not the answer.

Option C: While the transhumanists talk about how transhuman technology would make up for the frailness of the human body, there is no mention of whether they would be willing to have these devices in their own bodies, that too instead of human organs that are healthy. Hence Option C is the correct answer.

Option D: "Biotechnology will enable a union between humans and genuinely intelligent

computers and AI systems" and this is something the transhumanists strongly endorse, as can be gathered from the passage. Hence, Option D is incorrect. Choice (C)

Solution for question 17:

17. Sentence 1 references a previous sentence with 'this' European identity. It also draws a contrast between this identity and East Communism.

Sentence 2 is an independent sentence that hints at a rift – two factions.

Sentence 3 starts with a positive connector 'consequently' and talks about the origin of a Western Europe as opposed to an Eastern Europe.

Sentence 4 references a previous sentence (that, of course) to explain how that doesn't mean 'this' space was homogenous. So, the previous sentence must really be about a space that gives the impression of homogeneity.

If you notice, Sentences 2, 3 and 4 all try to draw a contrast between two groups. Only 4 talks about how even this group we are discussing earlier is not homogenous. So, it can be understood that 4 is the most downstream of all, the last sentence of the para.

Similarly, 2 begins the para for the simple logic that without 2, the sentences 1 and 3 don't make any sense at all. So, it boils down to 2134 and 2314. But 3 starts with 'consequently' but what happens in 1 (the virtues of this European identity) don't explain the content of 3 (identifying a Western Europe). Identifying the Western Europe comes before discussing the qualities/characteristics. So, 2314. Ans: (2314)

Solution for question 18:

18. The statements immediately preceding and following blank (1) talk about algal cells soaking up the sun's energy and transmuting that energy. Therefore, the two sentences form a mandatory pair and inserting the highlighted sentence between them would break the thought flow. Hence, the highlighted sentence does not fit into blank (1).

The statement immediately preceding blank (2) introduces chloroplasts and the statement immediately following blank (2) gives a little more insight regarding the background of the chloroplasts. The highlighted sentence mentions "bacterial tenants" and this would require a preceding sentence that talks about some kind of residence. But that is mentioned in the statement statements are connected. Hence, the highlighted sentence fits into blank (3).

The statement immediately preceding blank (4) talks about other bacteria that live inside other creatures. The highlighted sentence is in past tense - "the bacteria ***gave up***", and therefore, the statement preceding it should either be in past tense or allude to the past. This statement comes before blank (3) and not before blank (4). Hence, this is not the answer. Choice (C)

Solution for question 19:

19. Option A: This is incorrect, because the para doesn't say that operationalism 'outside' social sciences is largely dead. This implies that in social sciences it may still be relevant. But, that isn't true. Option A is not the answer.

Option B: This option gives us both the main ideas: that operationalism is largely dead, but not before it made a very powerful point about how to understand science – by looking at its practitioners. Hence, Option B is the answer.

Option C: Firstly, science is not 'all about what its practitioners do.' That is simply incorrect. We discussed practitioners because operationalism exhorts us to look at the practitioners to understand what science is about. Secondly, the option picks the less important trivial details of Skinnerian behaviourism and intelligence testing, over the more important point of operationalism's chief contribution – the powerful point it made about science. Option C is not the answer.

Option D: This option misrepresents the important point that operationalism made. It is to look at the practitioners and not to understand science from the viewpoint of its practitioners. Looking at them and understanding their viewpoint are entirely different things. Option D is not the answer. Choice (B)

Solution for question 20:

20. Sentence 1 is an independent sentence that explains how science is influenced by certain factors.

Sentence 2 is an independent sentence that explains the possibility of science not being bias-free ('it doesn't exist in a vacuum).

Sentence 3 is grammatically independent, but contextually we need to understand what observations are being spoken about here.

Sentence 4 offers an example – which seems to suggest that research showed male domination in

following blank (2). Therefore, the highlighted sentence has to follow that statement and hence, it does not fit into blank (2).
The statement immediately preceding blank (3) talks about bacteria that took up residence inside algae cells. "bacterial **tenants**" in the highlighted sentence refers to these bacteria taking up residence inside algae cells. "gave up their tough outer coats…" is the result of bacteria taking up residence inside algae cells. Therefore these two

primates.
We can understand that 2 is upstream to 1, although both are independent. That's because 2 is generic and only opens the possibility of bias in scientific method, whereas 1 comes down to greater details, bias because of culture. 3 is more an elaboration of what 2 introduces – how cultural factors influence observations. 4 is the example for the point 1 and 3 are making – the influence of existing culture (whatever it may be) on the

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research and observations being made. While the possibility of 12 block can be indulged, it must be noted that 13 make a block (theory and elaboration). And 2 is too vague to come after 13 block, because 13 block already discuss the essence of 2. Hence, 2 comes at the top of the para. So, 2134 is the order. Ans: (2134)

Solution for question 21:

21. The highlighted sentence talks about Bell's inequalities, something that is introduced in the second half of the paragraph. Therefore, the highlighted sentence cannot be part of blank (1) where the concept of Bell's inequalities has not been introduced yet. Hence, the highlighted sentence does not fit into blank (1).
The concept of Bell's inequalities has been introduced in the paragraph after the statement immediately following blank (2). Therefore, the highlighted sentence has to follow this statement and not precede it. Hence, the highlighted sentence does not fit into blank (2).
The statement immediately preceding blank (3) introduces Bell's inequalities, which are about certain conditions. In the statement immediately following blank (3), "these conditions" refer to "Bell's inequalities" mentioned in the statement immediately preceding blank (3). The former introduces the concept and the latter elaborates on it. Therefore, these two statements are connected

But another possibility is that economic behaviour has different outcomes in different situations (not necessarily in different systems). Also, normative economics is not just about the difference in outcomes. Option C is not the answer.
Option D: This option represents the essence of the para succinctly. Positive economics indeed tries to comprehend the economic systems (without being judgmental of the outcomes). Normative economics tries to understand the nature of outcomes of these systems (good or bad) and then figures out the right course of action – or a way to deal with the outcomes. Option D is the answer. Choice (D)

23. Option A: While most of this option is a reflection of the content of the para, there is one slip-up. The conformity or obedience that domination demands isn't ***rooted*** in respect (for the three parameters – customs, charm, rules/statutes/law). These are the three types of domination, but the author hasn't mentioned obedience coming out of respect for these three. Hence, Option A is not the answer.
Option B: This option exaggerates on several counts. There is no comparison in the para between fear and the three power structures (traditions, charisma, rules) of which the option mentions just two. Also, this option ends up taking about individuals who dominate rather than domination itself. Hence, Option B is easy to eliminate.

and inserting the highlighted sentence between them would interrupt the flow. Hence, the highlighted sentence does not fit into blank (3).

The statement immediately preceding blank (4) talks about Bell proving the conditions being violated in some setups. "all such theories" in the statement immediately following blank (4) requires a precedent which mentions theories. Without the highlighted sentence, there is no mention of any theories in the statement immediately preceding blank (4). Therefore, these two statements cannot form a pair. The highlighted sentence mentions theories and the statement immediately following blank (4) has to refer to the highlighted sentence. Hence, the highlighted sentence fits into blank (4).
Choice (D)

Solutions for questions 22 and 23:

22. Option A: The para mentions that positive economics attempts to understand economic systems. It does not judge outcomes, objectively or subjectively. Option A is not the answer.

Option B: Normative economics doesn't judge economic behaviour. It judges the outcomes of the behaviour in order to prescribe the course of action. Option B is not the answer.

Option C: While the option is true for a large part, the para doesn't indicate that there is a possibility of 'different types of economic behaviour'. Normative economics analyses outcomes of the economic behaviour. One possibility is indeed that different systems have different outcomes.

Option C: This option isn't the best. While there are no glaring errors, it misses out on the all-important characteristic of obedience that domination expects – wilful obedience. Instead, the option just mentions submission which changes the sense of meaning of the para. Hence, Option C is not the answer.

Option D: This is apt as it reflects all the ideas in the para – domination's demand for obedience that is willed, and the demand seeking legitimacy in the three parameters. Basically, those who hold power expect to dominate based on their personal charm, the law or rules or the tradition/custom. These three structures, the power holders believe, give their domination a sense of legitimacy or rightness. Option D is the answer. Choice (D)

Solution for question 24:

24. Sentence 1 is an independent sentence that explains the purpose of an essay as a medium for speaking out serious matters.

Sentence 2 has the personal pronoun 'they' but that refers to the second part of the sentence, the Victorian essayists. Similarly, the 'however' is not a contrast marker here. It is part of the expression 'however much'. So, this is an independent sentence that introduces a commonality between Victorian essayists.

Sentence 3 starts on a contrasting tone with 'but' followed by change. This sentence transports the tone from cultivated people to the not so cultivated ones.

Sentence 4 again refers to 'they' and it is clear that this is the commonality amongst the Victorian essayists- they wrote at greater length. This sentence also talks about 'high standards' of the readers, thereby, preceding the sentence that brings the transition from high standards to lower ones.

2 introduces the writers and the times (Victorian) which is needed for 1 – because it says 'it was worthwhile' (when – that's the obvious question). 4 has to appear before 1 as 'they' in 4 needs the Victorian essayists. Also, 1 continues the subject of 4. 3 comes in the end, talking about the

Given that B is the fourth tallest. The two persons adjacent to him must be the tallest or second tallest or the third tallest. Since the second tallest is not adjacent to B, the two persons standing adjacent to B must be the tallest and the third tallest. Hence, F and D must be the tallest and the third tallest, in any order.

From (i), A is standing adjacent to the third tallest person in the group. If F is the third tallest, A cannot be standing adjacent to him. Hence, D must be the third tallest and A must be standing to the immediate right of D. F must be the tallest person.

In positions 4, 5 and 6, there must be the fifth, sixth and seventh tallest persons, in any order. Further, for transition. So, 2413. Ans: (2413)

Difficulty level wise summary - Section I

Level of Difficulty	Questions
Very Easy	–
Easy	18
Medium	1, 2, 3, 4, 12, 13, 14, 21
Difficult	5, 6, 7, 8, 9, 10, 11, 15, 16, 17, 19, 20, 22, 24
Very Difficult	23

SECTION – II

Solutions for questions 1 to 5:

Let 1 to 7 represent the positions of the persons along the line such that 1 represents the position at the extreme left and 7 represents the position at the extreme right.

Given there are seven persons, and for three persons, both the persons standing adjacent to each of them are taller than he is. These three persons cannot be at any extreme end. Further, no two of these three persons can be next to each other (if they were, then for one of them, the given condition will not be satisfied). Hence, the three persons must be in alternate positions, i.e., at positions 2, 4 and 6. Further, these three persons cannot be the tallest or the second tallest (since there must be at least two persons taller than them).

The four persons in positions 1, 3, 5 and 7 will be taller than the three persons in the alternate positions. These four persons cannot be the shortest (since they have to be taller than the three persons adjacent to them).

From (iii), D is standing four places to the left of the second tallest person. The second tallest person can only be at 5 or 7 (for D to be standing four places to his left). However, D is not standing at any end. Hence, the second tallest person cannot be at 5. Hence, the second tallest person must be at 7 and D must be at 3.

From (ii), B is standing adjacent to the tallest person. Any person who is standing adjacent to the tallest person will be among the three who are shorter than both their neighbours. Hence, B can be at 2 or 4 or 6. However, from (ii), B is not standing adjacent to the second tallest person. Hence, B cannot be at position 6. From (iv), F is to the immediate left of B. Hence, B cannot be at 4 (because D will be at his immediate left). Hence, B must be at 2. From (iv), F must be at 1.

the positions 4 and 6, the two persons standing adjacent to these positions must be taller than the persons in 4 and 6. If the fifth tallest person is standing in position 4 or 6, then in position 5 there must be the sixth or the seventh tallest. But this is not possible because the two persons standing adjacent to the fifth tallest person will not be taller than him. Hence, the fifth tallest person cannot be at 4 or 6. The fifth tallest person will be in position 5. In positions 4 and 6, the sixth and seventh tallest persons must be present.

From (i), A is not the shortest person. Hence, A, at position 4, must be the sixth tallest. In position 6, there must be the shortest person.

From (v), at least three persons are taller than C and shorter than E. Hence, C can be fifth tallest or seventh tallest. If C is fifth tallest, from (v), E must be the tallest. However, this is not possible. Hence, C must be the seventh tallest. E must be the second tallest. G must be the fifth tallest.

The following table provides the order in which the seven persons are standing and their relative heights (with 1 representing the tallest and 7 representing the shortest person):

Order	1	2	3	4	5	6	7
Person	F	B	D	A	G	C	E
Height	1	4	3	6	5	7	2

1. A is the sixth tallest person. Choice (C)

2. F and G are standing two places away from the third tallest person (i.e., D). Among the two, G is given in the options. Choice (C)

3. There are two pairs of persons satisfying the given criteria: (B, D) and (A, G). Choice (D)

4. The maximum difference is for C and E, which is equal to 5. Choice (A)

5. Four persons are standing to the left of G. Among these four persons, three persons (F, B and D) are taller than G. Choice (B)

Solutions for questions 6 to 9:

From (iv), E's rank can be 5 or 6 or 7 and G's rank can be 1 or 2 or 3. Since E and G were not visited on the same day, E's rank and G's rank cannot be 5 and 1 OR 6 and 2.

Hence, E's rank must be 7 and G's rank must be 3.
From (i), C cannot be ranked 1st. B cannot be ranked 7th. Since he did not visit C on a Saturday, C cannot be ranked 4th or 7th. Hence, C can be ranked 2 or 5 or 6.
From (iii), D's rank cannot be 1. Since he did not visit D on Monday, D's rank cannot be 2 or 3 or 6.
Hence, D's rank can only be 4 or 5.
From (v), A's rank cannot be 1 or 2 or 3 or 6. Hence, A's rank can only be 4 or 5.
From (iii), A's rank must be 4 and D's rank must be 5.
From (v), A was ranked worse than F. Hence, F's rank can only be 1 or 2.
Also, B was ranked worse than C. Hence, B's rank can only be 1 or 2.
C's rank must be 6. From (ii), F's rank must be 2 and B's rank must be 1.
The following table provides the ranks of the seven restaurants and the day on which he visited each restaurant:

Rank	Restaurant	Day
1	B	Friday
2	F	Monday
3	G	Monday
4	A	Saturday
5	D	Friday
6	C	Monday
7	E	Saturday

6. Five restaurants were ranked better than C (which was ranked sixth). Choice (D)

7. Both B and F are ranked better than G.
Choice (D)

8. He must have visited the seven restaurants on Monday, Friday, Saturday, Monday, Friday, Saturday and Monday.
The second restaurant the he visits can be ranked 1 or 5 (since he visited this on Friday).
Let the second restaurant that he visited be ranked 5. The fifth restaurant must be ranked 1 (since the fifth restaurant was visited on a Friday).
The third restaurant must be ranked 7 (it cannot be ranked 4 as the difference between 5 and 4 is 1).
The first restaurant can be ranked 2 or 3.
The fourth restaurant can be ranked 2 or 3. However, this cannot be 2 as the fifth restaurant is ranked 1. Hence, the fourth restaurant is ranked 3 and the first restaurant must be ranked 2.
The sixth restaurant must be ranked 4 and the last restaurant must be ranked 6.
This is one possible case.
Let the second restaurant be ranked 1 and the fifth restaurant be ranked 5.
The first restaurant can be ranked 3 or 6.
The third restaurant can be ranked 4 or 7.
The fourth restaurant must be ranked 2 or 3. It cannot be ranked 6 because the fifth restaurant is ranked 5.
The sixth restaurant must be ranked 7. It cannot be ranked 4 because the fifth restaurant is ranked 5.
Hence, the third restaurant must be ranked 4. The fourth restaurant must be ranked 2.
The last restaurant cannot be ranked 6 (as the sixth restaurant is ranked 7). Hence, the last restaurant must be ranked 3 and the first restaurant must be ranked 6. This is another case.
These are the only two possible cases and they are presented below:

Order	Case 1		Case 2	
	Restaurant	Rank	Restaurant	Rank
1	F	2	C	6
2	D	5	B	1
3	E	7	A	4
4	G	3	F	2
5	B	1	D	5
6	A	4	E	7
7	C	6	G	3

He could have visited 2 or 5 restaurants before visiting A. Choice (D)

9. In both the cases, the required difference = 2.
Choice (B)

Solutions for questions 10 to 14:

Given that by the end of March 2019, all employees were skilled in all the three activities.
Since there are a total of 300 employees, and each employee, at the time of hiring, has exactly one skill, the employees will have a total of 300 skills among them. This cumulative number of skills should increase to 900 (300 employees × 3 skills) by the end of March 2019.
It is also given that every month, exactly 60 skills are taught (20 for Polishing, 10 for Buffing and 30 for Stitching). Hence, for imparting the additional 600 skills, a minimum of 10 months will be required. 10 months from June 2018 is March 2019.
∴ The employees were imparted the skills in the minimum possible time.
Also, for all the employees to learn exactly one new skill, it would require 5 months. Hence, by October 2018, every employee will have exactly two skills (since it is given that no employee can have three skills).
Also, for 10 months, every month 30 employees were taught Stitching. A total of 300 employees must have been taught Stitching in the 10 months. ∴ None of the employees would have been skilled in Stitching when they were hired.

For 10 month, 20 employees were taught Polishing, i.e., a total of 200 employees were taught Polishing. Hence, a total of 100 employees would be skilled in Polishing when they were hired.
Similarly, 100 employees were taught Buffing. Hence, 200 employees would be skilled in Buffing when they were hired.
In the month of October, every employee would have been skilled in two skills.
Since every month, 10 employees were taught Buffing and 20 employees were taught Polishing, a total of 30 employees would become skilled in both Buffing and Polishing every month. Hence, after 5 months, i.e., by the end of October, 150 employees would be skilled in both Buffing and Polishing.
Also, since 30 employees were taught Stitching, after 5 months, 150 employees will be skilled in Stitching. Among these 150 employees, some would have been skilled in Polishing and the rest would have been skilled in Buffing. In the next 5 months, all these employees will become skilled in all the three activities.
Since only 10 employees can learn Polishing, a total of 50 employees will be trained in Polishing in the next 5 months. Hence, only 50 employees cannot have the skill of Polishing by the end of October 2018. This must be the number of employees who are skilled in only Buffing and Stitching. From October onwards, every month, the number of employees who have only Buffing and Stitching as skills will reduce by 10.
Similarly, 20 employees are taught Buffing every month and in 5 months, a total of 100 employees are taught Buffing. In October 2018, only 100 employees cannot have Buffing as a skill. This must be the number of employees who are skilled in only Polishing and Stitching. From October onwards, every month, the number of employees who have only Polishing and Stitching as skills will reduce by 10.
The number of employees who have Polishing and Buffing as skill at the end of October 2018 (150 employees) will reduce by 30 each month until the end of March 2019.
The table below provides the number of employees with different skills in each month. In the table, P represents Polishing, B represents Buffing and S represents Stitching.

Beginning of	P	B	S	Only P+B	Only P+S	Only B+S	P+B+S
June 2018	100	200	0	0	0	0	0
November 2018	0	0	0	150	50	100	0
December 2018	0	0	0	120	40	80	60
January 2019	0	0	0	90	30	60	120
February 2019	0	0	0	60	20	40	180
March 2019	0	0	0	30	10	20	240
April 2019	0	0	0	0	0	0	300

10. At the end of October 2018, 50 employees were skilled in Polishing but not in Buffing.
Ans: (50)

11. At the end of December 2018, 90 employees were not skilled in Stitching but were skilled in Buffing.
Ans: (90)

12. From the above solution, for a_{xyz} and b_{xyz} to be equal, every month, 15 employees who know Polishing must be taught Stitching, while 15 employees who know Buffing must be taught Stitching from June to October.
Since P + S can become a maximum of 50, this can happen only in 3 months from June to October.
employees from P move to P + B and 20 from B move to P + B (this cannot be changed).
The following table provides one way in which this can be done:

Beginning of	P	B	P + S	P + B	B + S
June	100	200	0	0	0
July	90	150	0	30	30
August	80	100	0	60	60
September	70	50	0	90	90
October	40	20	20	120	100
November	0	0	50	150	100

Hence, the maximum number of months in which a_n and b_n can be equal is 3. Choice (D)

13. For the number of employees skilled only in Polishing to be greater than the number of employees skilled only in Buffing, we need to maximize the number of people moving from B to B + S and minimize the number of employees moving from P to P + S. Also, every month, 10

From the table, we can see that in two months, a_{xyz} can be greater than b_{xyz}. Choice (C)

14. From June 2018 to October 2018, every month, 10 employees from P will move to P + B, while 20 employees from B will move to P + B.
Hence, P + B will increase by 30 every month.

Also, some employees from P, say x, will move from P to P + S.
And, 30 – x employees from B will move to B + S.
Hence, P + S and B + S will together increase by 30 every month.
Also, the total number of employees who can move from P to P + S over the five months is 50 and the total number of employees who can move from B to B + S over the five months is 100 (from the values calculated for the end of October).
In this question, it is given that x = a_{xyz}.
Since we need to maximize the number of months for which a_{xyz} = 0, we can assume that the 50 employees who move from P to P + S do so in the minimum number of months. If 30 employees are taught Stitching in one month and 20, in another month, then, in the first five months, in three months, a_{xyz} can be 0. After October till March, in every month 10 employees who were skilled in Polishing will be taught Stitching. Hence, there cannot be any month in the last five months where a_{xyz} = 0.
∴ The maximum number of months in which a_{xyz} = 0 is 3. Choice (A)

Solutions for questions 15 to 18:

There are six numbers, and each child was given two numbers. Since each number was given to not more than two children, each number must have been given to exactly two children.

If B was given 15, the other number must be 18. C would have been given 18 and 5. This is a possible case.
If C was given 15, the other number must be 6. B must have been given 5 and 6 (the remaining numbers which are not given twice). This case is not possible as the product for B was not 30.
If B was given 9, then the other number must be 30. C must have been given 30 and 3. A must have been given 3 and 5. But the product for A is not 15. This case is also not possible.
Hence, only one case is possible and that is presented below:

Child	Numbers
A	9, 15
B	15, 18
C	18, 5
D	7, 9
E	5, 10
F	7, 10

15. The numbers given to A and C are distinct. Choice (D)

16. The sum of the twelve numbers = (5 + 7 + 9 + 10 + 15 + 18) × 2 = 128 Ans: (128)

From the given information, to find the numbers given to each child, we need to split the given product into two factors both greater than 1. By observing the numbers, we can see that 63 has the least number of possibilities of splitting into two factors.
The numbers given to D can be 3 and 21 or 7 and 9.
If the numbers given to D are 3 and 21, there must be one other number which is a multiple of 21. However, there is no such number. Hence, these cannot be the two numbers.
Therefore, the numbers given to D must be 7 and 9.
One more child must be given 7. By observing the other products, we can see that only one other number is a multiple of 7, i.e., 70. Hence, F must be given 7 and 10.
One more child must be given 9. Looking at the products, this can be A or B or C.
Since F is given 10, one more child must be given 10. This can be B or C or E.
Since there are many cases here, we can look at the other smaller number among the given numbers, i.e., 50, given to E.
E can be given 2 and 25 or 5 and 10. However, there is no other product which is a multiple of 25. Hence, E cannot be given 2 and 25.
E must have been given 5 and 10.
Among A, B and C, one child must be given 9 and one must be given 5.
If C was given 9, then the other number must be 10. This is not possible as the number 10 was given to two children already (E and F). Hence, C was not given 9.
If A was given 9, then the other number must be 15. Between B and C one person must be given 15.

17. For three children, A, B and C, the sum of the numbers is greater than 20. Ans: (3)

18. The difference between the two numbers is the highest for C. Choice (B)

Solutions for questions 19 to 22:

From (iii), the number of boys in the school in 2018 was 130. Hence, the number of girls in the school in 2017 was also 130. From (iii), the number of girls in 2015 was 140. Hence, the number of boys in 2016 was also 140.
From (ii), the number of boys in 2019 was 160. Hence, the number of girls in 2018 was 160. The total number of students in 2018 is 290.
From (i), the number of students in 2017 must be 140 + 30 = 170. Since there are 130 girls in 2017, there must be 40 boys in 2017. Hence, there must be 40 girls in 2016. In 2016, the number of students will be 180.
From (ii), the number of students in one of the years is 320. This cannot be 2019 because there are 160 boys in 2019 and if the total is 320, then there will be an equal number of boys and girls in the school, which is not possible.
Hence, the number of students in 2015 must be 320. Since there are 140 girls in 2015, there must be 180 boys in 2015.
From (i), the number of students in 2019 must be 300. Hence, the number of girls in the school in 2019 must be 140.
The following table provides the above information:

Year	Boys	Girls	Total
2015	180	140	320
2016	140	40	180
2017	40	130	170
2018	130	160	290
2019	160	140	300

19. The difference between the number of boys in the school in the year 2015 and the number of girls in the school in the year 2016 = 180 – 40 = 140 Choice (D)

20. The total number of students in the school in 2017 = 170. Choice (B)

2. Let the two integers be a, b.
$ab - a - b = -42$ ----- (1) OR
$ab - a - b = 42$ ----- (2)
(1) $\Rightarrow ab - a - b + 1 = -41$
$\Rightarrow (a - 1)(b - 1) = -41$
$\Rightarrow a - 1 = -1$ and $b - 1 = 41$
i.e., $a = 0$, $b = 42$ or $a - 1$
$= -41$ and $b - 1 = 1$ i.e., $a = -40$, $b = 2$
$\therefore$ {a, b} can be {0, 42} or {–40, 2}
(2) $\Rightarrow ab - a - b + 1 = 43$
$\Rightarrow (a - 1)(b - 1) = 43$
$\Rightarrow a - 1 = 1$, $b - 1 = 43$
i.e., {a, b} = {2, 44} or $a - 1 = -1$, $b - 1 = -43$
i.e., {a, b} = {0, –42}
$\therefore$ Four pairs of numbers are possible. Ans: (4)

3. As the thickness is uniform, weight is proportional to area.

21. In 2018, the number of students increased to 290 from 170, in the previous year. Hence, at least 120 students must have enrolled in 2018.

Ans: (120)

22. The total number of students increased from 290 to 300. However, the number of boys increased from 130 to 160. Hence, at least 30 boys must have enrolled in the school. The minimum number of students who must have enrolled = 30.

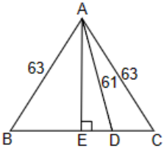
Ans: (30)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	6, 7, 15, 16, 17, 18, 19, 20
Medium	8, 9, 21, 22
Difficult	1, 2, 3, 4, 5, 10, 11, 14
Very Difficult	12, 13

SECTION – III

Solutions for questions 1 to 24:

1.



We drop a perpendicular from A to BC which meets BC at E where BE = EC (Since ΔABC is an isosceles triangle)

Now $AE^2 = AC^2 - EC^2$ and $AE^2 = AD^2 - ED^2$

Equating AE^2, we get

$AC^2 - EC^2 = AD^2 - ED^2$

$AC^2 - AD^2 = EC^2 - ED^2$

$63^2 - 61^2 = (EC - ED)(EC + ED)$

$(2)(124) = (DC)(BD)$ (Since EC = BE)

= 248. Choice (C)

For any equilateral triangle, the percentage of area of maximum possible circle is the same.

Area of the equilateral triangle with side s is $\frac{\sqrt{3}}{4}s^2$

In an equilateral triangle, the radius of the maximum possible circle (in-circle) is one-third the length of the median of the triangle.

$\therefore$ radius = $\frac{1}{3}\left(\frac{\sqrt{3}}{2}s\right) = \frac{1}{2\sqrt{3}}s.$

$\therefore$ Area of circle = $\pi\left(\frac{s}{2\sqrt{3}}\right)^2 = \frac{\pi s^2}{12}$

$\therefore$ Required percentage = $\dfrac{\left(\frac{\pi s^2}{12}\right)}{\left(\frac{\sqrt{3}}{4}s^2\right)}$

$\cong \frac{3.14}{12} \times \frac{4}{1.732} \cong 60.4\%$ Choice (C)

4. Tap A was open for the first 15 minutes, before tap C is opened.

$\therefore$ Part of the tank filled by tap A = $\frac{15}{40}$.

Before C is opened, Tap 'B' was opened for 5 minutes.

$\therefore$ Part of the tank filled by tap B = $\frac{5}{60}$.

After 'C' is opened remaining part of the tank left to be filled = $1 - \left(\frac{15}{40} + \frac{5}{60}\right) = \frac{65}{120}$. When A, B, C operate together, fraction of the tank that gets filled in one minute.= $\frac{1}{40} + \frac{1}{60} + \frac{1}{100} = \frac{15+10+6}{600} = \frac{31}{600}$.

$\therefore$ Time taken to fill the remaining part of the tank

$= \dfrac{\frac{65}{120}}{\frac{31}{600}} = \frac{65 \times 5}{31} = \frac{325}{31} = 10\frac{15}{31}$ minutes.

Alternate solution:

Let the three pipes together be opened for x minutes.
Then A was opened for $x + 15$ minutes, B for $x + 5$ minutes and c for x minutes.

$$\therefore \frac{x+15}{40}+\frac{x+5}{60}+\frac{x}{100}=1$$

$\Rightarrow x = 10\frac{15}{31}$ Choice (D)

5. $\log_{\sqrt{2}} 2^2 + \log_{3\sqrt[3]{3}} 3^3 + \ldots + \log_{50\sqrt[50]{50}} 50^{50}$

$= 2\log_{\sqrt{2}} 2 + 3\log_{\sqrt{3}} 3 + 4\log_{\sqrt{4}} 4 + \ldots\ldots 50\log_{\sqrt{50}} 50$

$= \frac{2}{(1/2)}\log_2 2 + \frac{3}{(1/3)}\log_3 3 + \frac{4}{(1/4)}\log_4 4 \ldots\ldots \frac{50}{(1/50)}\log_{50} 50$

$= 2^2 + 3^2 + 4^2 + \ldots 50^2$

$\left(\log_{a\sqrt[a]{a}} a^a = \frac{a}{(1/a)}\log_a a = a^2\right)$

$= \frac{50(50+1)(2\times 50+1)}{6} - 1^2 = 42925 - 1 = 42924$

Ans: (42924)

6. Whenever the price of an item, which has initially ₹x, is increased by p% and then decreased by p%, the final price of the item would be

$x\left(1-\frac{p^2}{100}\right)$ and the net reduction in price after consecutive increase and decrease is equal to $\frac{xp^2}{100}$

Given,

$\therefore \frac{xp^2}{100} = 16 \Rightarrow \frac{p^2}{100} = \frac{16}{x}$ --- (1)

If the same operation is performed once again the final price would be

$$x\left(1-\frac{p^2}{100}\right)\left(1-\frac{p^2}{100}\right) = 368.64$$

$$\Rightarrow x\left(1-\frac{16}{x}\right)\left(1-\frac{16}{x}\right) = 368.64$$

Substituting the given choices, we can say
x = ₹400 Choice (A)

7. $N = 364n^5 = 2^2 \times 7^1 \times 13^1 \times n^5$
As 2, 7 and 13 are factors of N, so N has at least three prime factors.
Again, it is given that N has 182 factors
$182 = 2^1 \times 7^1 \times 13^1$
Thus, N must have exactly three prime factors.
$N = (p_1)^1 (p_2)^6 (p_3)^{12}$
and No. of factors of N = (1 + 1) (6 + 1) (12 + 1) = 182.
$\therefore n = 2^x \times 7^y$ or $2^x \times 13$ ($\because$ 2's power in N is more than 1 and 2 must be present in n)

$\therefore n^5 = 2^{5x} \times 7^{5y}$ or $2^{5x} \times 13^{5y}$

We observe that x must be 2 and y must be 1
$\therefore N = 2^2 \times 7^1 \times 13^1 (2^2 \times 7)^5 = 2^{12} \times 7^6 \times 13^1$ or
$N = 2^2 \times 7^1 \times 13^1 (2^2 \times 13)^5 = 2^{12} \times 13^6 \times 7^1$
Thus, N can assume two values. Ans: (2)

8. $t_n = n(n+2) = n^2 + 2n$
$S_n = \sum n^2 + \sum 2n$

$$= \frac{n(n+1)(2n+1)}{6} + 2\left(\frac{n(n+1)}{2}\right)$$

$$S_{15} = \frac{15\times 16\times 31}{6} + 15 \times 16$$

$$= 15 \times 16\left(\frac{37}{6}\right) = 5 \times 8 \times 37 = 1480$$

Alternative Solution:

Since the sum required is only upto 15 terms and the series is very simple, we could directly (and carefully) put down all 15 terms and calculate the required sum as 3 + 8 + 15 + 24…….255 = 1480.
Choice (A)

9. Let the number which 952 divides be N.
So, N = 952q + 124 where q is the quotient.
Rearranging, N = 68(14q + 1) + 56.
When 68 divides N, the quotient would be 14q + 1 and the remainder will be 56. Ans: (56)

10. $(x + y + z)^2 \geq 0$
$\Rightarrow x^2 + y^2 + z^2 + 2(xy + yz + zx) \geq 0$

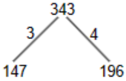
$$\Rightarrow xy + yz + zx \geq \frac{-(x^2+y^2+z^2)}{2}$$

$\therefore xy + yz + zx \geq -2$
Again $(x - y)^2 + (y - z)^2 + (z - x)^2 \geq 0$
$2(x^2 + y^2 + z^2) - 2(xy + yz + zx) \geq 0$
$\Rightarrow xy + yz + zx \leq x^2 + y^2 + z^2$
$\Rightarrow xy + yz + zx \leq 4$
Therefore, the range for $xy + yz + zx$ is [– 2, 4]

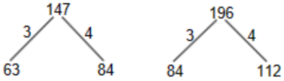
Alternative Solution:

Let $x = y = z = k$.
Then $3k^2 = 4 \Rightarrow xy + yz + zx = 3k^2 = 4$.
This eliminates choices (A) and (B).
Now, let $x = 0$, $y = \sqrt{2}$ and $z = -\sqrt{2}$, then $xy + yz + zx$ $= -2$.
This eliminates choice (C). Hence, choice (D) must be the answer. Choice (D)

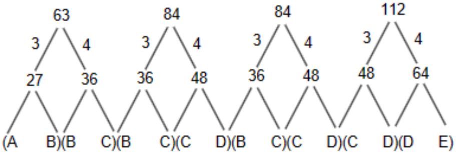
11. When the rope of length cut in the ratio 3:4



Now, these pieces are cut in ratio 3:4



Repeating it for 2 more times



We will observe that, we will have

1 piece of length $27 \times \frac{3}{7} = 11.57$ (A)

4 pieces of length $27 \times \frac{4}{7}$ (or) $36 \times \frac{3}{7} = 15.42$ (B)

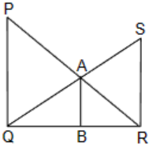
6 pieces of length $36 \times \frac{4}{7}$ or $48 \times \frac{3}{7} = 20.57$ (C)

4 pieces of length $48 \times \frac{4}{7}$ or $64 \times \frac{3}{7} = 27.42$ (D)

1 piece of length 36.57 (E)

After arranging in ascending order, the seventh piece will be of length 20.57m. Choice (B)

12.



As ΔPQR and ΔABR are similar, $\frac{AB}{PQ} = \frac{RB}{RQ}$ ---- (1)

Alternative Solution:

If PQ = RS, i.e., i = j = 1 (say), then AB should be $\frac{1}{2}$, irrespective of the value of k (from symmetry). Plugging in i = j = 1 in the choices, only choice (C) gives AB = $\frac{1}{2}$, irrespective of k.

Choice (C)

13. $6 \phi 8 = (6)(8) \phi 6$ [$\because 6 < 8$] $= 48 \phi 6 = \frac{2(48)-6}{48+6}$

As ΔSRQ and ΔABQ are similar, $\frac{AB}{SR} = \frac{BQ}{RQ}$ ---- (2)

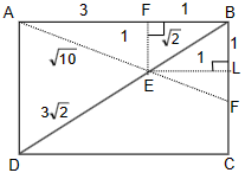
Adding (1) and (2), $AB\left(\frac{1}{PQ} + \frac{1}{SR}\right) = \frac{RQ}{RQ}$

$$AB = \frac{PQ \times SR}{PQ + SR} = \frac{IJ}{I + J}$$

$[\because 48 > 6]$

$= \frac{90}{54} = \frac{5}{3}$. Choice (D)

14.



E is a point on the diagonal BD.

$\therefore$ it is equidistant from sides AB and BC.

$EF \Rightarrow EL = 1$

$BF = 1$ and $AF = 3$

$\therefore BE = \sqrt{2}$ and $AE = \sqrt{3^2 + 1^2} = \sqrt{10}$

$BD = \text{diagonal} = 4\sqrt{2}$, and $BE = \sqrt{2}$

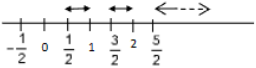
$\therefore DE = 3\sqrt{2}$

X travels $3\sqrt{2}$ km (DE) in 1 hr.

$\therefore$ he will travel $\sqrt{2}$ km (EB) in 1/3 hour = 20 minutes.

Alternative Solution:

If the product of 5 numbers is non negative, then either 1, 3 or all 5 of them are non-negative.



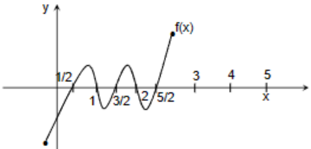
The interval shown satisfies (1), i.e.

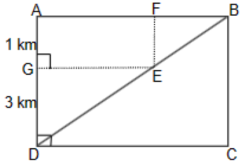
$\frac{1}{2} \le x \le 1$ or $\frac{3}{2} \le x \le 2$ or $x \ge \frac{5}{2}$

So, the possible single digit numbers are 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9. So, a total of 9 single digit numbers.

Alternate Solution 2:

The roots of equation (1) above are 1/2, 1, 3/2, 2, 5/2. The plot of the equation (1) can easily be drawn as below.





Draw EG parallel to CD.
Δles GED and ABD are similar.

$$\therefore \frac{GD}{DE} = \frac{AD}{DB} = \frac{AD - GD}{DB - DE} = \frac{AG}{EB}$$

$$\Rightarrow \frac{AG}{GD} = \frac{EB}{DE} = \frac{1}{3}$$

$\therefore$ Time taken by X to cover EB = $\frac{1}{3} \times 1$ hour
= 20 minutes. Choice (B)

15. We need to check for $x = 0, 1, 2, \ldots\ldots.9$.
If $x = 0$, then the given expression
$= (-1)(-2)(-3)(-4)(-5)$, which is negative.
If $x = 1$, $(2x - 2) = 0$, if $x = 2$, $(2x - 4) = 0$ and if $x > 2$ (i.e., for $x = 3, 4, \ldots..9$) we see that all the terms of the expressions are positive.

Alternative Solution 1:
Given that
$(2x - 1)(2x - 2)(2x - 3)(2x - 4)(2x - 5) \geq 0$

$$\Rightarrow \left(x - \frac{1}{2}\right)(x - 1)\left(x - \frac{3}{2}\right)(x - 2)\left(x - \frac{5}{2}\right) \geq 0 \quad \ldots(1)$$

It can clearly be observed that except for $x = 0$, f(x) is non-negative for all whole numbers.
Hence, f(x) ≥ 0, for $x = 1, 2, \ldots\ldots$ up to 9.
Ans: (9)

16. The five boys can be arranged amongst themselves in 5! i.e., 120 ways. Now among P, Q and R, there are 3! arrangements possible and each of these 6 possibilities are equally likely to occur.
1. P Q R
2. P R Q
3. Q P R
4. Q R P
5. R P Q
6. R Q P

Of these only the first case satisfies the criterion asked. So $\frac{120}{6}$ i.e., 20 ways are there in which the given condition will be satisfied.

Alternative Solution:

If the order P, Q and R is fixed, then one of the remaining two (say S) can come in four possible places as shown below

__ P __ Q __ R __.

The last person (T) can then be arranged in any of five possible places (similar to that above).
Hence, 4 × 5 = 20. Choice (B)

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17. Interest accrued in three years = (1.375 – 1) = 0.375

$= \frac{3}{8}^{\text{th}}$ of the sum.

Hence, interest per annum = $\frac{1}{8}$ of sum for interest to equal twice the sum, number of years required

$= \frac{2}{\left(\frac{1}{8}\right)} = 16$. Ans: (16)

$2(a_1 + a_3 + a_5 + \ldots\ldots + a_{15}) = 6^8 - 4^8$
$\therefore a_1 + a_3 + a_5 + \ldots\ldots.. + a_{15}$

$= \frac{6^8 - 4^8}{2} = 807040$ Ans: (807040)

22. Number of chocolates is less than 500.
As 7 chocolates will be remaining after distribution chocolates to 11 students
The number of chocolates will be of form $11k_1 + 7$
Similarly, it can be of the form $13k_2 + 12$
$11k_1 + 7 = 13k_2 + 12$

18. $x^2 + y^2 - 6y - 12x + 28 = 0$
$x^2 - 12x + 36 + y^2 - 6y + 9 + 28 = 36 + 9$
$(x-6)^2 + (y-3)^2 = 17$

The above curve is a circle with centre at (6, 3) and radius $\sqrt{17}$ units. The least distance between the point and circle

= |Distance between the point and centre – Radius of the circle|

Distance between the point (4, 0) and the center (6, 3) is $\sqrt{(6-4)^2+(3-0)^2} = \sqrt{2^2+3^2} = \sqrt{13}$

Radius = $\sqrt{17}$ units

$\Rightarrow$ least distance = $\sqrt{17} - \sqrt{13}$ units.

Choice (B)

19. Given expression is
$x^2 y^3 = 256 \Rightarrow x^2 y^3 = (16)^2 (1)^3$ or $(2^2)(4^3)$
As x and y are positive and $x > y$, x = 16 and y = 1.
$\therefore 3x - 2y = 3(16) - 2 = 46$ Choice (A)

20. Let the number of kites bought be 210 (LCM of 5, 7 and 6)

Number of white kites = $\frac{3}{5}(210) = 126$

Number of yellow kites = 210 – 126 = 84

Number of kites sold = $\frac{3}{7}(210) = 90$

Number of kites unsold = 120

Number of yellow kites sold = $\frac{1}{6}(84) = 14$

Number of white kites sold = 90 – 14 = 76
Number of white kites unsold = 126 – 76 = 50

	White	Yellow	Total
	126	84	210
Sold	76	14	90
Unsold	50		120

$\therefore \frac{\text{No of unsold white kites}}{\text{No of unsold kites}} = \frac{50}{120} = \frac{5}{12}$

Choice (D)

21. It is given that,
$(4 + x + x^2)^8 = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots\ldots + a_{16}x^{16}$
Putting $x = 1$, we get
$6^8 = a_0 + a_1 + a_2 + \ldots\ldots a_{15} + a_{16}$ -------- (1)
Again putting $x = -1$, we get
$4^8 = a_0 - a_1 + a_2 - a_3 + \ldots\ldots a_{15} + a_{16}$ ----- (2)
Subtracting equation (2) from equation (1), we get,

$k_1 = k_2 + \frac{2k_2+5}{11}$

As k_1 is an integer, $2k_2 + 5$ must be divisible by 11
$\therefore k_2 = 3$
Smallest number = 51
Number of chocolates will be of form $(11 \times 13)k + 51 = 143k + 51$
Let N be number of chocolates and r be the chocolates remaining when divided among students:

if k = 0, N = 51, r = Rem $(\frac{51}{15}) = 6$

if k = 1, N = 194, r = 14
if k = 2, N = 337, r = 7
if k = 3, N = 480, r = 0
∴ Nirvaan cannot be left with 12 chocolates.

Choice (C)

23. Let the present ages of Ajay and Vijay be 7x and 3x.
Thirteen years ago, their ages were 7x – 13, 3x – 13, respectively.
$\frac{7x-13}{3x-13} = \frac{9}{2}$
$14x - 26 = 27x - 117$
$13x = 91$
$x = 7$
Present ages are 49 and 21 years. After sixteen years, their ages would be 65 and 37 years.

Choice (A)

24. N marbles are distributed among P children, where P ≠ 1 or N. Hence any number "N" below hundred which has at least one factor "P" other than the number itself and which satisfies the condition $2 < N + P < 100$, can be a possible value of N. First, from the first 99 natural numbers, we eliminate all primes below 100. 99 – 25 = 74. Further 1, 99, 98 and 95 also do not qualify, since for these numbers the smallest values of P, do not satisfy $2 < N + P < 100$. Hence, N can take 74 – 4 or 70 values. Ans: (70)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	13
Easy	5, 8, 9, 12, 17, 19, 20, 23
Medium	1, 3, 4, 6, 14, 15, 16
Difficult	2, 7, 10, 11, 18, 21, 22, 24
Very Difficult	–

Done
Tip: Scroll slowly for the best results